

संघ २०८१

लोक सेवा आयोग

नेपाल इञ्जिनियरिङ्ग सेवा, सिभिल समूह अन्तर्गत हाइवे, स्यानिटरी र हाइड्रोपावर उपसमूह, राजपत्रांकित तृतीय श्रेणी (प्राविधिक) पदको
प्रतियोगितात्मक लिखित परीक्षा

बिषय: Technical Subject | पत्र: द्वितीय | पूर्णाङ्क: १०० | समय: ३ घण्टा | मिति: २०८१/१०/१२

Section-A (30 Marks)

1. For a soil sample with maximum dry density 1.86 t/m^3 at 15% water content and of specific gravity 2.7, calculate degree of saturation, air content and percentage of voids at the maximum dry density. Assume any relevant data if you need. 5

Answer:

a) Given Data:

- Maximum Dry Density (ρ_d) = 1.86 t/m^3
- Water Content (w) = 15% = 0.15
- Specific Gravity (G_s) = 2.7
- Density of water (ρ_w) = 1.0 t/m^3 (Assumed)

b) Calculations:

- Void Ratio (e):

Using the formula $\rho_d = \frac{G_s \cdot \rho_w}{1+e}$

$$1.86 = \frac{2.7 \times 1}{1+e} \Rightarrow 1+e = \frac{2.7}{1.86} = 1.4516$$

$$\therefore e = 0.4516$$

- Degree of Saturation (S):

Using $S \cdot e = w \cdot G_s$

$$S = \frac{0.15 \times 2.7}{0.4516} = 0.8968$$

$$\therefore S = 89.68\%$$

- Air Content (a_c):

$$a_c = 1 - S = 1 - 0.8968 = 0.1032$$

$$\therefore a_c = 10.32\%$$

- Percentage Air Voids (n_a):

$$\text{First, find porosity (n): } n = \frac{e}{1+e} = \frac{0.4516}{1.4516} = 0.3111$$

$$n_a = n \cdot a_c = 0.3111 \times 0.1032 = 0.0321$$

$$\therefore n_a = 3.21\%$$

2. Describe in brief the advantages and limitations of pre-stressed concrete. 5

Answer:

Introduction:

Pre-stressed concrete is a form of concrete where internal stresses are introduced (usually via high-tensile steel tendons) to counteract the tensile stresses resulting from external loads.

Advantages:

- Longer Spans: Enables the construction of longer clear spans compared to RCC, making it ideal for bridges in Nepal's hilly terrain.
- Slender Sections: Since the entire concrete section is effective in resisting loads, smaller and more aesthetic sections can be used, reducing self-weight.
- Crack Control: The permanent compressive force keeps cracks closed under service loads, improving durability and protecting reinforcement from corrosion.
- High Shear Resistance: The "cable profile" often provides an upward component of force that reduces diagonal tension (shear).
- Economy in High Loads: For heavy-duty structures like industrial floors or bridges, it becomes more economical than RCC.

Limitations:

- High Material Cost: Requires high-strength concrete (M35+) and high-tensile steel (strands/wires), which are more expensive than standard RCC materials.
- Specialized Labor: Requires skilled technicians for tensioning and grouting operations.
- Equipment Cost: Needs specialized jacks, anchors, and sometimes heavy casting beds.

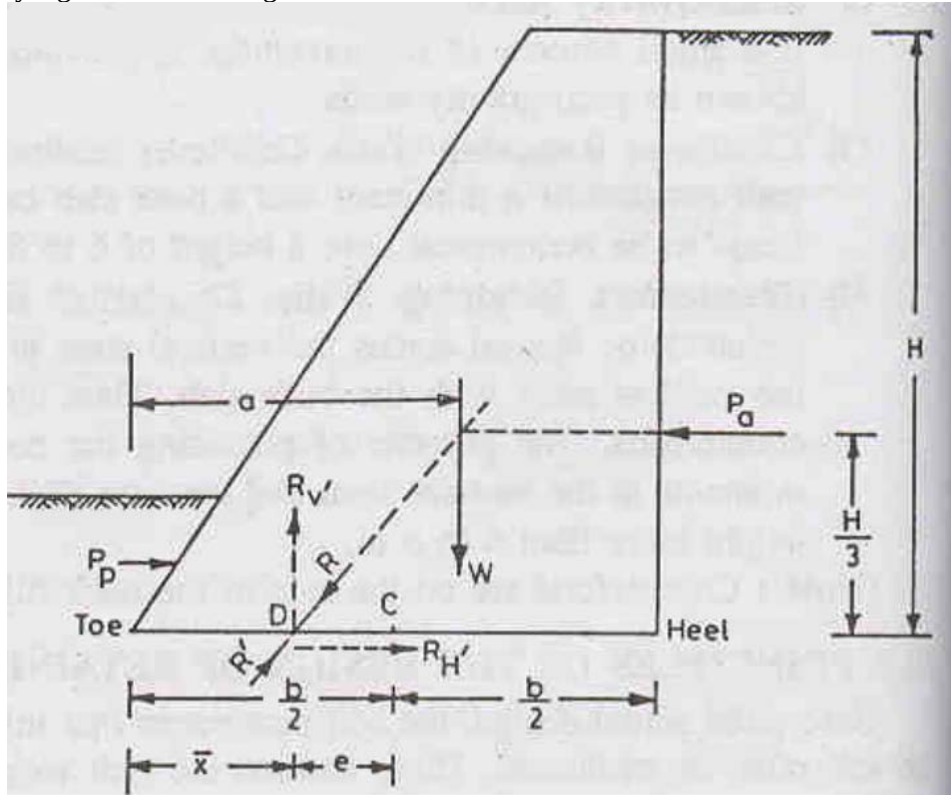
- Complex Design: Design calculations (losses of pre-stress, bursting tension) are more complex than conventional RCC.

3. Using suitable formula and diagrams, carry out the stability analysis of an earth retaining wall. 10

Answer:

Stability analysis ensures the wall remains safe against external forces (Lateral Earth Pressure, Self-weight, Surcharge).

a) Stability Against Overturning:



The wall must not tip over its toe.

$$FOS_{overturning} = \frac{\sum M_{Resisting}}{\sum M_{Overturning}} \geq 2.0 \text{ (for permanent structures)}$$

$M_{Resisting}$ = Weight of wall $\times$ Distance from toe

$M_{Overturning} = P_a \times \frac{H}{3}$ (for triangular pressure)

b) Stability Against Sliding:

The wall must not move horizontally.

$$FOS_{sliding} = \frac{\mu \cdot \sum V}{P_a} \geq 1.5$$

μ = Coefficient of friction between base and soil.

$\sum V$ = Total vertical force (Self-weight + Surcharge).

c) Stability Against Bearing Capacity Failure:

The pressure at the base must not exceed the Allowable Bearing Capacity (q_{all}) of the soil.

$$q_{max/min} = \frac{\sum V}{B} \left(1 \pm \frac{6e}{B} \right)$$

Condition: $q_{max} < q_{all}$ and $q_{min} \geq 0$ (To avoid tension at the base).

d) Check for Eccentricity (Middle Third Rule):

To ensure no tension develops at the base, the resultant force must act within the middle third of the base.

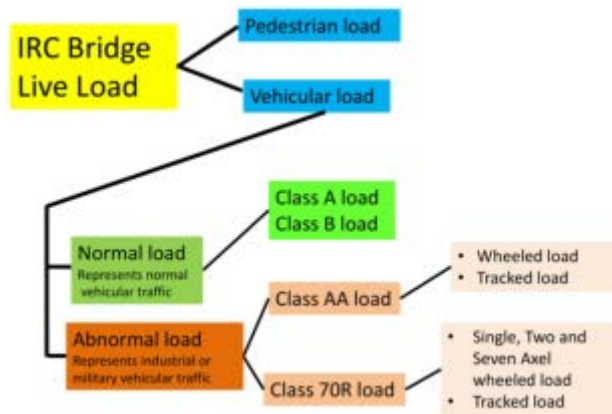
$$e = \frac{B}{2} - \frac{\sum M_{net}}{\sum V} \leq \frac{B}{6}$$

4. Explain the different considerations for structural design of bridges focussing on live load, impact load, wind load and centrifugal force. In addition, provide the design principles for solid slab bridges. 5+5=10

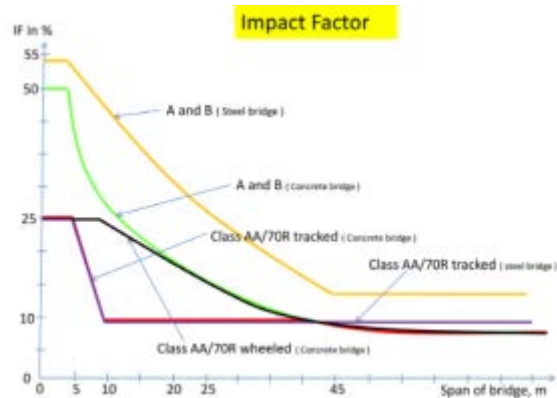
Answer:

A. Structural Design Considerations (Forces):

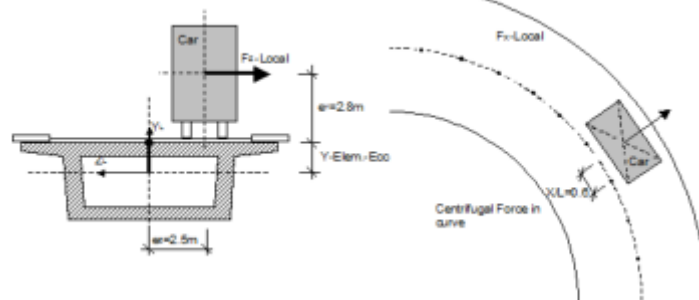
- Live Load: Represents the weight of vehicles. In Nepal, Nepal Bridge Standard (NBS) or IRC (Class AA, 70R, Class A/B, Pedestrian) loadings are used. It is placed in the most eccentric position to produce maximum stress.



- Impact Load: Accounts for the dynamic effect of moving vehicles. It is taken as a fraction of the live load ($I = \frac{A}{B+L}$), depending on the span and bridge type.



- Wind Load: Significant for long-span or high-elevation bridges. It acts on the exposed area of the structure and moving vehicles.
- Centrifugal Force: If the bridge is on a horizontal curve, this force acts horizontally through the center of gravity of the vehicle. Calculated as $C = \frac{WV^2}{gR}$.



B. Design Principles for Solid Slab Bridges:



Solid slab bridge

- Applicability: Economical for short spans (typically up to 8--10 m).
- Effective Span: Taken as the clear span plus the effective depth or the distance between centers of bearings.
- Bending Moment: Calculated based on standard wheel loads; transverse distribution of live loads is often analyzed using Pigeaud's Curves or the Effective Width Method.
- Reinforcement: Main reinforcement is provided along the span; distribution reinforcement (at least 20% of main steel) is provided perpendicular to the span to handle temperature and shrinkage.
- Camber: Slight upward camber is provided to offset long-term deflection.

Section-B (25 Marks)

- How can one estimate depth and frequency of irrigation for an irrigation command area for irrigation scheduling? Write down. 5

Answer:a) 1. Depth of Irrigation (d):

This is the amount of water needed to bring the soil moisture from its current level (usually the Management Allowed Depletion - MAD) back to Field Capacity (FC).

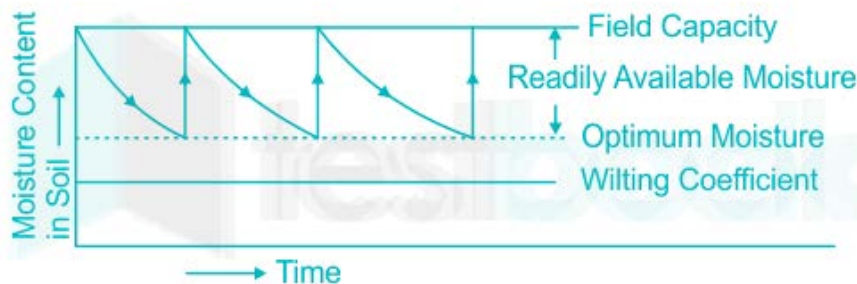
$$d = \frac{(W_{fc} - W_{om})}{100} \cdot \gamma_d \cdot D$$

Where:

- W_{fc} = Moisture content at Field Capacity (%)
- W_{om} = Optimum moisture content before irrigation (%)
- γ_d = Dry unit weight of soil
- D = Root zone depth of the crop

2. Frequency of Irrigation (f):

The interval between two consecutive irrigations. It depends on how fast the crop uses water (Consumptive Use).



$$f = \frac{d_w}{Cu}$$

Where:

- d_w = Net depth of water available for use ($d \times \text{Efficiency}$)
- Cu (or ET_c) = Daily Consumptive Use (Evapotranspiration) of the crop.

Scheduling Principle: In Nepal, scheduling is critical during the "Critical Growth Stages" (e.g., flowering/fruiting) to prevent yield loss.

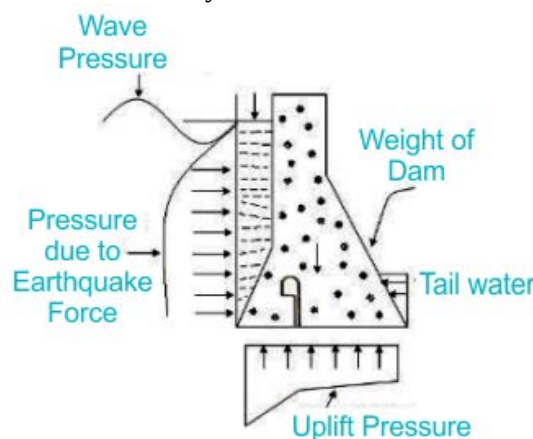
6. **What are the advantages of concrete gravity dams? Show the different forces acting on gravity dams with a neat sketch. How do you analyze the stability of concrete gravity dams against sliding and overturning? Elaborate. 2+3+5=10**

Answer:**Advantages:**

- Durability:** Highly resistant to decay and requires low maintenance.
- Safety:** Best suited for high-head projects and can withstand heavy floods over the crest (if designed as an overflow section).
- Adaptability:** Can be constructed on any site where a strong rock foundation is available.

Forces Acting on a Gravity Dam:

Gravity dams are massive structures that rely on their own weight to resist external loads. The forces acting on them are categorized based on their role in stability.

**A. Primary Stabilizing Force**

- Self-Weight of the Dam:** This is the most critical resisting force. It acts vertically downward through the dam's center of gravity and provides the necessary moment to counteract overturning.

- b) **Vertical Component of Water Pressure (on an inclined upstream and downstream face):** If the upstream face and downstream face is inclined, the weight of the water column resting on this slope acts vertically downward, contributing to the total downward force that stabilizes the dam.
- c) **Tailwater Pressure (on the downstream face):** The pressure exerted by water stored on the downstream side acts horizontally towards the upstream, helping to stabilize the dam against the overturning forces from the reservoir.

B. Major Destabilizing (Overturning) Forces

- a) **Water Pressure (Hydrostatic):** The major external force pushing against the upstream face. It varies linearly with depth, forming a triangular pressure distribution that is zero at the surface and maximum at the base.
- b) **Uplift Pressure:** Water seeping through the dam's foundation or joints exerts an upward force. This effectively reduces the net weight of the dam, making it more vulnerable to sliding or toppling.

C. Secondary & Exceptional Forces

- a) **Earthquake Forces:** In seismic zones, these are crucial. They include horizontal inertia forces and hydrodynamic pressure (the "sloshing" effect of reservoir water).
- b) **Silt Pressure:** Accumulation of silt on the upstream face exerts additional horizontal and vertical pressure.
- c) **Wave Pressure:** Wind blowing over the reservoir generates waves that impact the upper portion of the dam.
- d) **Ice Pressure:** In cold climates, expanding ice sheets can exert a horizontal thrust at the reservoir level.
- e) **Wind Pressure:** Generally considered only for very high dams when the reservoir is empty.

Stability Analysis:

- a) Against Overturning:
Calculated about the toe. The Resisting Moments (mostly due to self-weight) must exceed the Overturning Moments (water and uplift pressure).

$$FOS_{OT} = \frac{\sum M_R}{\sum M_O} > 1.5 \text{ to } 2.0$$

- b) Against Sliding:
The horizontal water pressure must be resisted by the friction between the dam base and the rock.

$$FOS_{sliding} = \frac{\mu(\sum V - U)}{P} > 1.0$$

Note: For higher safety, the Shear Friction Factor (SFF) is also checked.

7. **The mean monthly flow of a typical Nepalese River is given below. The headwater level is 1,200 m , whereas Tailwater level is 800 m and can be considered as constant. Assuming a design discharge as 50 m³/s, total head loss as 3%, overall efficiency of the plant as 90%, compute firm and secondary energy produced by the plant. What is the contribution of dry energy (Dec-May) to the total generation? Assume any relevant data you need. 10**

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
11	10	13	20	31	63	134	164	111	56	31	17	55

Answer:

a) Data Analysis:

- Gross Head (H_g) = 1200 – 800 = 400 m
- Head Loss = 3% $\Rightarrow$ Net Head (H_n) = 400 $\times$ 0.97 = 388 m
- Design Discharge (Q_d) = 50 m³/s
- Overall Efficiency (η) = 0.90
- Power Formula: $P = \eta \cdot \rho \cdot g \cdot Q \cdot H_n = 0.9 \times 9.81 \times Q \times 388 = 3425.6 \cdot Q$ (in kW)

b) Power/Energy Calculation Table (Summary):

- Minimum Flow (Q_{min}): 10 m³/s (February)
- Firm Power (P_f): Based on $Q_{min} = 10$ m³/s
 $P_f = 3425.6 \times 10 = 34,256$ kW = 34.256 MW
- Total Annual Energy: $\sum(P_{monthly} \times \text{Hours in month})$ where $P_{monthly}$ is capped at P_{design} (using $Q = 50$ m³/s).
- Firm Energy (E_f): $P_f \times 8760$ hrs = 34.256 $\times$ 8760 = 300,082 MWh

- Secondary Energy (E_s): Total Energy – Firm Energy. (This occurs when $Q > 10 \text{ m}^3/\text{s}$, up to $50 \text{ m}^3/\text{s}$).

c) **Dry Energy Contribution (Dec–May):**

Sum the energy generated in Dec, Jan, Feb, Mar, Apr, and May.

$$\text{Contribution\%} = \frac{\text{Dry Energy}}{\text{Total Annual Energy}} \times 100$$

(Note: In Nepal's context, Dry Energy is the most valuable and usually commands a higher PPA rate).

Section-C (25 Marks)

8. Introduce International Civil Aviation Organization (ICAO) and briefly describe its roles. 2+3=5

Answer:

Introduction:

ICAO is a specialized agency of the United Nations established in 1944 (Chicago Convention). It manages the administration and governance of international civil aviation. Nepal became a member in 1960.

Roles of ICAO:

- Standardization:** Develops **Standards and Recommended Practices (SARPs)** for flight safety, security, and efficiency (19 Annexes).
- Safety & Security:** Coordinates international efforts to reduce aviation accidents and prevents unlawful interference (terrorism).
- Environmental Protection:** Minimizes the impact of aviation on the environment through noise and emission standards.
- Licensing:** Sets standards for the licensing of flight crews, air traffic controllers, and aircraft maintenance engineers.

Technical Cooperation: Provides assistance to member states (like Nepal) in developing their civil aviation infrastructure and regulatory framework.

9. Importance of highway drainage systems and state the different types of drainage in hill roads. 4+6=10

Answer:

Importance of Highway Drainage:

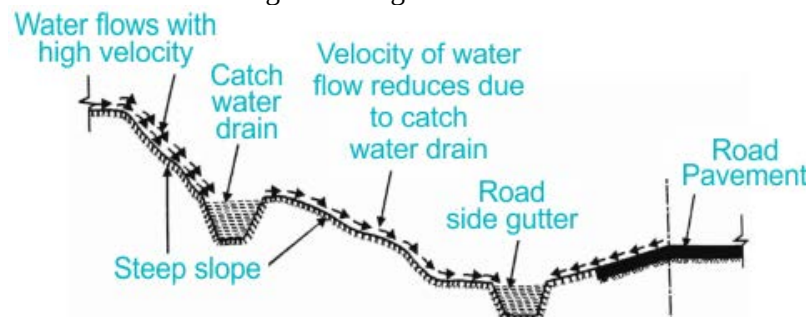
- Pavement Life Extension:** Prevents the softening of the subgrade and stripping of bitumen in flexible pavements, while reducing mud pumping in rigid pavements.
- Safety Enhancement:** Rapidly removes water to prevent hydroplaning (skidding) and improves visibility and tire grip for high-speed traffic.
- Structural Stability:** Prevents erosion of side slopes and embankments. In hilly areas, it is critical to reduce the risk of landslides caused by water-induced pore pressure.
- Economic Maintenance:** By preventing early failures like potholes, cracks, and ruts, it significantly reduces long-term maintenance and repair costs.
- Frost Action Prevention:** Minimizes damage caused by frost heave in cold climates where trapped water expands upon freezing.

Types of Drainage in Hill Roads:

Hill roads require a combination of systems to handle steep gradients and heavy runoff.

A. Surface Drainage Systems

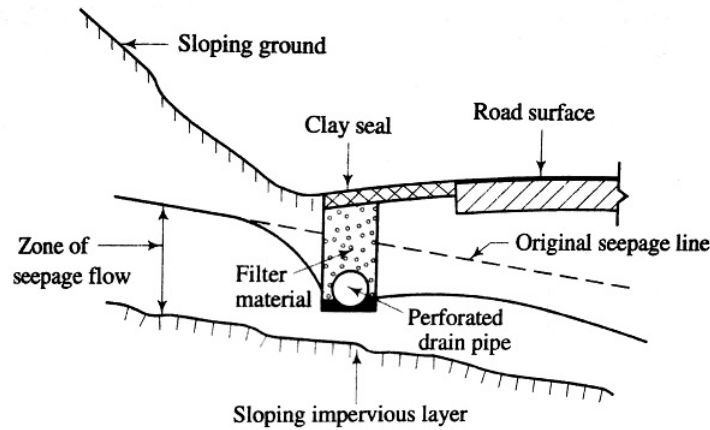
Designed to collect and divert water flowing over the ground and road surface.



- Camber (Cross-Slope):** The transverse slope of the road that directs water to the side drains.
- Side Drains:** Parallel to the road, usually V-shaped, trapezoidal, or **saucer-shaped** on the hill side to collect runoff.
- Catch Water Drains (Interceptor Drains):** Located at higher elevations on the hill slope to intercept runoff before it reaches the road and lead it to **cross drains**.
- Chutes (Sloping Drains):** Paved channels that carry water safely down steep embankments to prevent erosion.

B. Sub-Surface Drainage Systems

Essential for controlling **groundwater** and seepage that can weaken the roadbed.

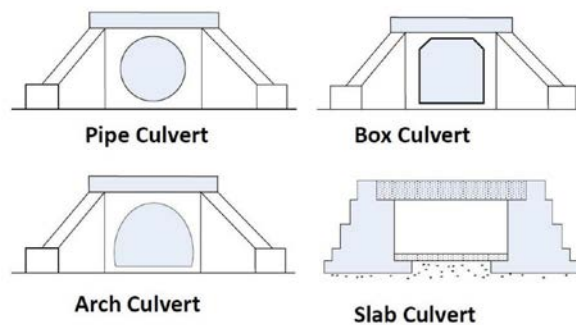


- a) **Deep Side Drains:** Used to intercept seepage and lower the **groundwater table** below the subgrade level (typically 1.2m).
- b) **Capillary Cut-offs:** Layers of granular material or **impermeable membranes** (e.g., bituminous layers) that stop moisture from rising into the subgrade via capillary action.
- c) **Horizontal Drains:** Perforated pipes installed into the hill slope to drain internal water and stabilize the slope.

C. Cross-Drainage (CD) Works

Used wherever the road intersects a natural stream or water flow.

- a) **Culverts:** Structures with a span less than 6m (Pipe, Slab, Box, or Arch types).



- b)
- c) **Scuppers:** Small openings through parapet walls or curbs to allow water to discharge directly to the valley side.
- d) **Causeways (Submersible Bridges):** Used for seasonal streams; includes **low-level (flush)** and **high-level (vented)** versions.
- e) **Aqueducts and Inverted Siphons:** Used to carry water over or under the road when typical culverts are not feasible.

D. Special Structures

- a) **Catch Pits:** Structures provided at the entrance of cross drains to collect debris and sediment, preventing **clogging**.
- b) **Energy Dissipators:** Built at the outlets of steep drains to reduce the velocity of water and prevent **scouring** at the base.

10. Give an introduction of 'Flexible' and 'Rigid' pavements with their suitability. How do you calculate different loads on highway pavement? Explain the construction method of concrete pavement.

3+3+4=10

Answer:

Flexible Pavements:

- **Introduction:** These pavements consist of a mixture of bituminous materials and aggregates. They are called "flexible" because the entire pavement structure "bends" or deforms slightly under traffic loads. The load is transferred from the top layer to the bottom through grain-to-grain contact of the aggregates.
- **Suitability:** Flexible pavements are suitable for areas with low to medium traffic and where the subgrade soil is subject to settlement, as they can accommodate slight base deformations without cracking. They are also preferred when the initial construction budget is limited.

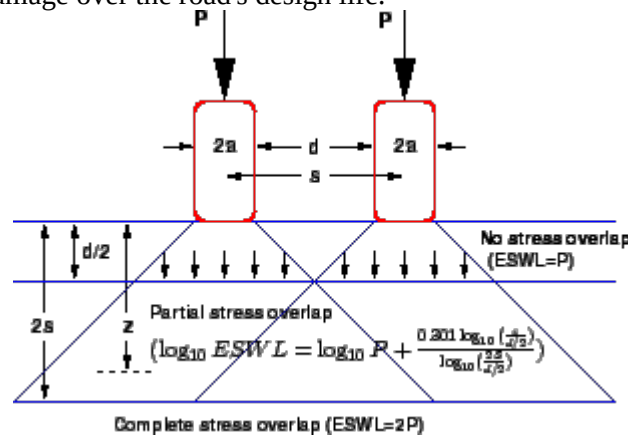
Rigid Pavements:

- **Introduction:** These are constructed using Portland Cement Concrete (PCC) slabs. They possess high flexural strength, allowing the slab to act as a structural beam and bridge over minor irregularities in the subgrade. Loads are distributed over a wider area through "slab action".

- **Suitability:** Rigid pavements are ideal for heavy traffic corridors, such as highways and industrial roads, due to their high load-carrying capacity and long design life (often 30–40 years). They are also highly suitable for areas prone to oil spills or waterlogging, as concrete is resistant to these elements

Calculation of Loads on Highway Pavement

Calculating pavement loads involves converting various vehicle types and weights into a standardized unit to determine the cumulative damage over the road's design life.



- **Standard Axle Load:** In design, different vehicle loads are converted into a Standard Axle Load, typically defined as a single axle with dual wheels carrying 80 kN (approx. 8.16 tonnes).
- **Vehicle Damage Factor (VDF):** This is a multiplier used to convert the number of commercial vehicles into the number of standard axle repetitions. It is often calculated using the Fourth Power Law:

$$VDF = \left(\frac{\text{Axle Load of Vehicle}}{\text{Standard Axle Load}} \right)^4$$

- **Cumulative Standard Axles (CSA):** The total design traffic (N) is calculated using the formula:

$$N = \frac{365 \times [(1 + r)^n - 1]}{r} \times A \times D \times F$$

Where:

- o A : Initial traffic after completion (Commercial Vehicles Per Day).
- o r : Annual traffic growth rate.
- o n : Design life in years.
- o D : Lane distribution factor.
- o F : Vehicle Damage Factor (VDF).
- **Equivalent Single Wheel Load (ESWL):** Since heavy vehicles use dual wheels, engineers calculate the ESWL—the single wheel load that would cause the same stress at a specific depth as the dual-wheel assembly

Construction Method of Concrete Pavement (Rigid):

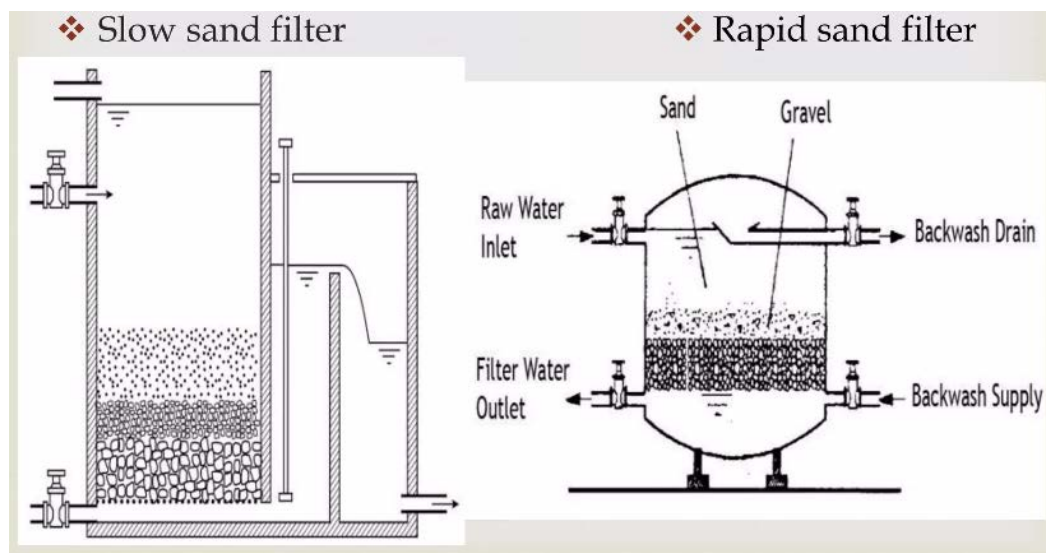
- Preparation of Subgrade/Sub-base:** The foundation is compacted; usually, a Dry Lean Concrete (DLC) base is provided.
- Formwork:** Fixed forms (steel/timber) are set to the required grade and alignment.
- Watering and Oiling:** Sub-base is moistened, and forms are oiled for easy removal.
- Placing Reinforcement:** If required (for JRC), steel mesh/dowels/tie bars are placed.
- Concrete Mixing and Placing:** Concrete (M30+) is mixed and spread evenly.
- Compaction & Finishing:** Use of screed/needle vibrators followed by "floating" and "texturing" (brooming) for skid resistance.
- Curing:** Done using water ponding or curing compounds for at least 14–28 days.
- Joint Cutting and Sealing:** Transverse/longitudinal joints are cut and filled with sealant to prevent water ingress.

Section-D (20 Marks)

11. Differentiate clearly between slow and rapid sand filter. Also, explain cleaning and maintenance of slow sand filter. 6+4=10

Answer:

Comparison Table:



Parameter	Slow Sand Filter	Rapid Sand Filter
Filtration Rate	Very slow, typically 0.1-0.3 m/h (100-200 L/hr/m ²).	Much faster, typically 4-21 m/h (4000-7200 L/hr/m ²).
Filtration Mechanism	Primarily a biological action via the <i>schmutzdecke</i> layer on the surface.	Primarily physical straining and adsorption throughout the bed, often aided by chemical coagulation.
Pre-treatment	Generally not required for raw water with low turbidity (<50 NTU).	Essential , requiring coagulation, flocculation, and sedimentation.
Cleaning Method	Scraping off the top 1.5-3 cm layer of sand manually.	Backwashing with treated water and compressed air (automatic).
Cleaning Frequency	Infrequent , every 1 to 3 months.	Frequent , typically every 1 to 2 days.
Area Required	Large surface area needed due to slow rate.	Small surface area needed due to high rate.
Bacterial Removal	Highly efficient , removing 90-99% of bacteria.	Less efficient, removing about 80-90% of bacteria, so disinfection is a must.
Sand Size	Finer sand with an effective size of 0.25-0.35 mm.	Coarser sand with an effective size of 0.45-0.70 mm.
Operation	Simple to operate, requiring less skilled supervision.	More complex operation, requiring well-trained personnel for monitoring.
Initial Cost	High installation cost but lower operation cost.	Lower installation cost but higher operation cost.

Cleaning and Maintenance of SSF:

- Frequency: Usually every 1–3 months when the "Head Loss" reaches the limit (0.7--1.2 m).
- Process: The top 1.5–3 cm layer of sand (called *Schmutzdecke*) is carefully scraped off.
- Resanding: After several scrapings, when the sand depth drops below the minimum (approx. 0.6 m), new clean sand is added.
- Ripening: After cleaning, the filter is run to waste for 24–48 hours until the biological layer reforms.

12. Why is environmental study necessary before executing an infrastructure development project? Process of EIA study and its approval. 3+7=10

Answer:

Necessity of Environmental Study:

- Legal Compliance:** Mandatory under Nepal's **Environmental Protection Act (EPA 2076)** and **EPR 2077**.
- Sustainability:** Ensures the project is environmentally sound and socially acceptable.
- Risk Mitigation:** Identifies potential negative impacts (landslides, deforestation) and provides mitigation measures.
- Public Participation:** Incorporates concerns of local communities, reducing future conflicts.

EIA Process and Approval in Nepal:

- Screening:** Determining if the project needs an IEE or EIA (based on thresholds).
- Scoping & ToR:** Identifying key issues and getting the **Terms of Reference (ToR)** approved by the Ministry.
- Baseline Data Collection:** Physical, biological, socio-economic, and cultural surveys.

- **Impact Prediction:** Assessing the magnitude, extent, and duration of impacts.
 - **Mitigation Measures:** Following the hierarchy: **Avoidance** → **Minimization** → **Compensation**.
 - **Public Hearing:** Mandatory 15-day public notice and hearing in the project area.
 - **Report Preparation & Submission:** Submitted to the concerned Ministry and then to the **Ministry of Forest and Environment (MoFE)**.
 - **Review & Approval:** MoFE reviews the report (often via a technical committee) and grants approval within a statutory timeframe.
-